



CMR COLLEGE OF ENGINEERING & TECHNOLOGY
(UGC AUTONOMOUS)

B.Tech II Semester Supplementary Examinations December-2025

Course Name: ODEs and Multivariable Calculus

(Common for all Branches)

Date: 30.12.2025 AN

Time: 3 hours

Max.Marks: 70

(Note: Assume suitable data if necessary)

PART-A

Answer all TEN questions

Each question carries TWO marks.

10x2=20M

1. Solve $ydx - xdy = a(x^2 + y^2)dx$ 2 M
2. State Newton's law of cooling. 2 M
3. Solve $(D^2 + 2D + 1)y = e^{-x}$ 2 M
4. Define wronskian of two functions and give an example. 2 M
5. Find the limits after changing the order of integration $\int_0^b \int_0^{\sqrt{b^2 - y^2}} f(x, y) dx dy$. 2 M
6. Using triple integral, find the volume of a rectangular box whose length is 6 feet, breadth is 5 and height is 4 feet. 2 M
7. Find a unit normal vector to the surface $z = x^2 + y^2$ at $(-1, -2, 5)$. 2 M
8. If $\vec{F} = y(ax^2 + z)\vec{i} + x(y^2 - z^2)\vec{j} + 2xy(z - xy)\vec{k}$ is solenoidal then find a . 2 M
9. State Stroke's theorem. 2 M
10. Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = x^2\vec{i} + y^2\vec{j}$ and the curve $y = x^2$ in the xy - plane from $(0, 0)$ to $(1, 1)$. 2 M

PART-B

Answer the following. Each question carries TEN Marks.

5x10=50M

- 11.A). i) Solve $(1 + y^2)dx = (\tan^{-1}y - x)dy$ 5M
 ii) Solve $x^2p^2 - 2yp + ax = 0$. 5M

OR

11. B). The temperature of the surrounding air is 20°C . The temperature of a hot body reduces from 100°C to 80°C in 10 minutes. What will be the temperature of the body after 20 Minutes? When will be the temperature 40°C ? 10M

12. A). Solve the differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 8e^{2x} \sin 2x$. 10M

OR

12. B). Solve $(D^2 + 4)y = \tan 2x$ by variation of parameters. 10M

13. A). Change the order of integration in $\int_0^1 \int_{x^2}^{2-x} xy \, dx dy$ and hence evaluate the same. 10M

OR

13. B). Evaluate $\iiint_V dxdydz$ where V the finite region of space is formed by the planes $x = 0, y = 0, z = 0$ and $2x + 3y + 4z = 12$. 10M

14. A). Show that $\nabla^2[f(r)] = f''(r) + \frac{2}{r}f'(r)$ where $r = |\vec{r}|$. 10M

OR

14. B). Find constants a, b, c so that the vectors $\vec{A} = (x + 2y + az)\vec{i} + (bx - 3y - z)\vec{j} + (4x + cy + 2z)\vec{k}$ is irrotational. Also find ϕ such that $\vec{A} = \nabla\phi$. 10M

(P.T.O.)

15. A). Verify Stoke's theorem for $\vec{F} = y^2\vec{i} - 2xy\vec{j}$ taken round the rectangular bounded by $x = \pm b, y = 0, y = a$. 10M

OR

15. B). Use divergence theorem to evaluate $\iint (xi + yj + z^2k) \cdot nds$ where S is the surface bounded by the cone $x^2 + y^2 = z^2$ in the plane $z = 4$. 10M
